



Pelita Bangsa
Academy

Interacting with IDRX Stablecoin

BUILDing with IDRX Stablecoin on Lisk Bootcamp

Hi, I'm Jason Yapri from Indonesia



Swiss German University

S.Kom. | Information Technology



Fachhochschule Südwestfalen

B.Eng. | Information Technology



University of Nottingham

M.Sc. | Entrepreneurship



University of Nicosia

MOOC | NFTs and the Metaverse

5+ years

Experience in the IT industry

Working Experience

(Startup, Corporate, Freelance | Germany, China, Singapore, Indonesia)



Teaching Experience



Pelita Bangsa Academy

PT Pelita Teknologi Bangsa

Mall of Indonesia, Lantai 2, Unit 2F-A7B,
Jalan Boulevard Barat Raya No. 1,
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pelitabangsa.co.id



[instagram.pelitabangsa.co.id](https://www.instagram.com/pelitabangsa.co.id)



[linkedin.pelitabangsa.co.id](https://www.linkedin.com/company/pelitabangsa.co.id)



[whatsapp.pelitabangsa.co.id](https://wa.me/pelitabangsa.co.id)

”

Upskill dan Reskill Talenta Teknologi
Indonesia di bidang Blockchain sehingga
kalian bisa bersaing secara global.



SESI

01

**Introduction to Lisk Blockchain
& IDRX Stablecoin**

02

Smart Contract Development on Lisk

03

Interacting with IDRX Stablecoin

HARI INI

SESI

04 Building Frontend UI for Web3 DApp

**Extra
Class**

Introduction to Xellar

05 Integrating Xellar Kit with Smart Contract

06 Brainstorming Web3 Project Ideas

16 Apr

17 Apr

23 Apr

30 Apr

Agenda

**Goal of this
Bootcamp**

Materi

Hands On

Q&A

Wrap Up

Foto Bersama

Recommendation



Physical Notebook



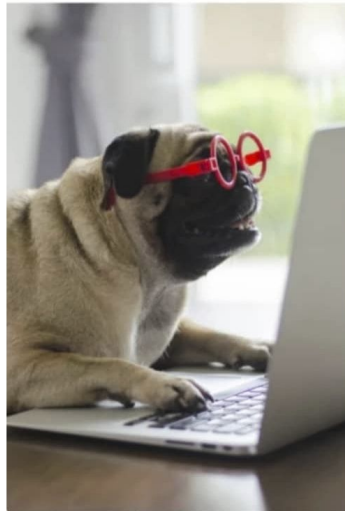
<https://notion.com>

Be Active

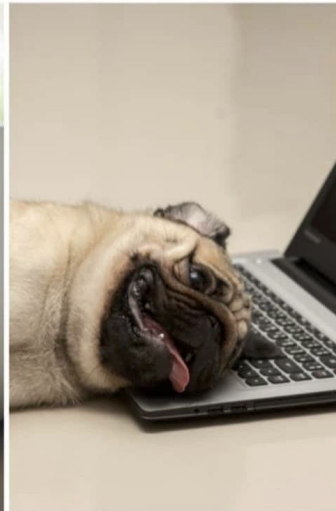


Open Your Camera

Zoom meeting,
with video



Zoom meeting,
audio only



A

Goal of this Bootcamp



Pelita Bangsa Academy Lisk 1000 IGNITE
BlockDevId X IDR X Xellar

* ♦ *

BUIDLing with IDRX Stablecoin on Lisk Bootcamp!

* ♦ *

 GRATIS!

LIVE via Zoom

→ Swipe untuk daftar

The goal of this bootcamp:

- Learn about Lisk Blockchain Technology
- Come up with Web3 Application Ideas
- #BuildOnLisk Blockchain
- Participate Lisk Builders Challenge Round 1
- Build successful Web3 Startup !!



Fund Tracking Management





Lisk

BlockDevId



IDRX



Xellar



LISK

Builders
Round 1
Challenge





Project Requirement:

1. Smart Contract deployed on **Lisk Sepolia Testnet** or **Lisk Mainnet** and verified on **Blockscout Explorer**
2. Integrate your dApp with **IDRX Stablecoin** on **Lisk Mainnet** or **Lisk Sepolia Testnet**
3. Build your dApps with **Xellar Kit**
4. Live Project URL

IDRX Address on Lisk Mainnet: 0x18Bc5bcC660cf2B9cE3cd51a404aFe1a0cBD3C22

IDRX Address on Lisk Sepolia Testnet: 0xD63029C1a3dA68b51c67c6D1DeC3DEe50D681661

Prize:

3,000 USD (1st Winner)

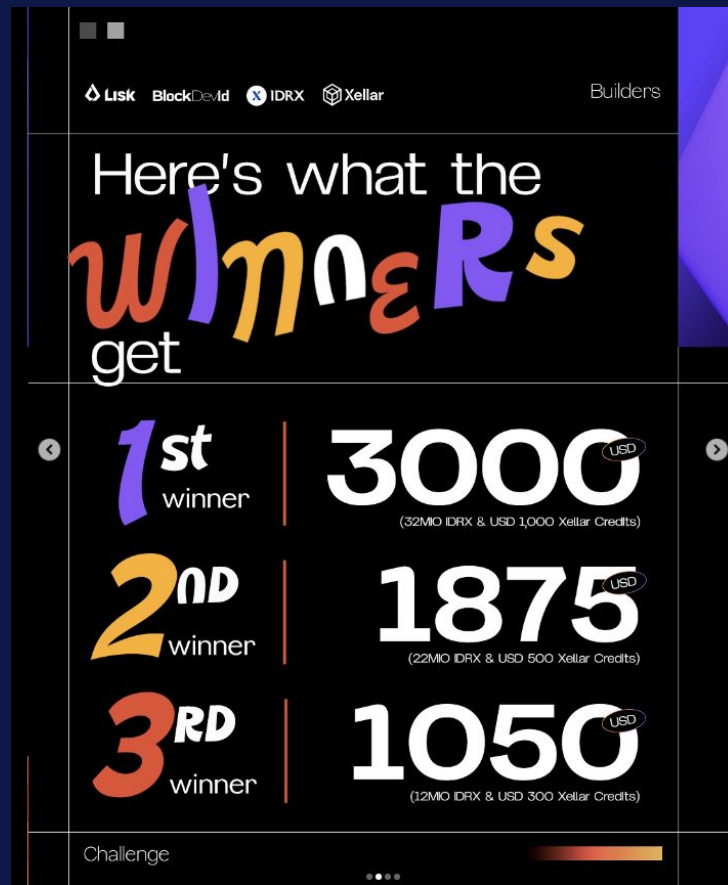
(32 Million IDR^X & USD 1,000 Xellar Credits)

1,875 USD (2nd Winner)

(22 Million IDR^X & USD 500 Xellar Credits)

1,050 (3rd Winner)

(12 Million IDR^X & USD 300 Xellar Credits)

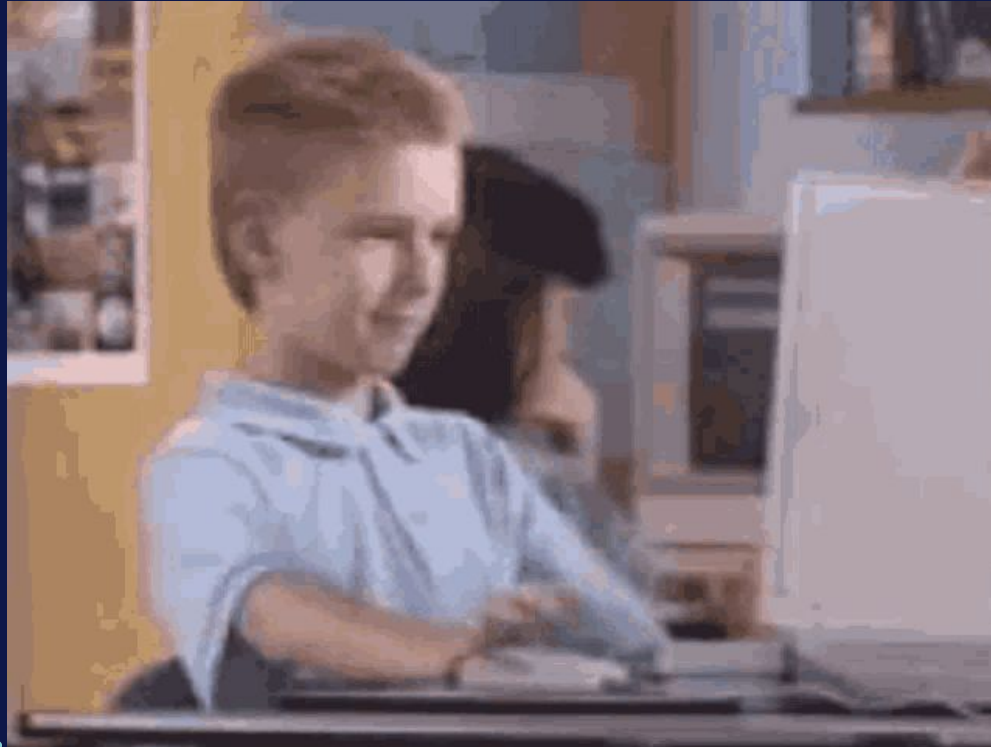


Here's what the winners get

1 st winner	3000 ^{USD} (32MIO IDR ^X & USD 1,000 Xellar Credits)
2 nd winner	1875 ^{USD} (22MIO IDR ^X & USD 500 Xellar Credits)
3 rd winner	1050 ^{USD} (12MIO IDR ^X & USD 300 Xellar Credits)

Challenge

Not bad





Prize:

8 Million IDR (Most Favorite Project)

3 Million IDR (SocMed Challenge Winner #1)

3 Million IDR (SocMed Challenge Winner #2)

The screenshot displays the Lisk Builders challenge interface. At the top, the Lisk logo and other project logos (BlockDevId, IDR, Xellar) are visible. The main content area is divided into three sections. The first section, titled 'More CHANCES to win', features a large, colorful, stylized word 'CHANCES' in red, blue, and yellow. The second section, titled 'Most Favorite Project', shows a prize of '8MIO IDR' (500 USD) for the 'Social Media Challenge'. The third section, titled 'Most Engaging & Creative', shows a prize of '3MIO IDR' (187.5 USD/each) for 'For 2 Winners'. The bottom of the interface shows a 'Challenge' tab and a progress indicator.

Lisk BlockDevId IDR Xellar Builders

More **CHANCES** to win

Most Favorite Project **8MIO** IDR (500 USD)

Social Media Challenge

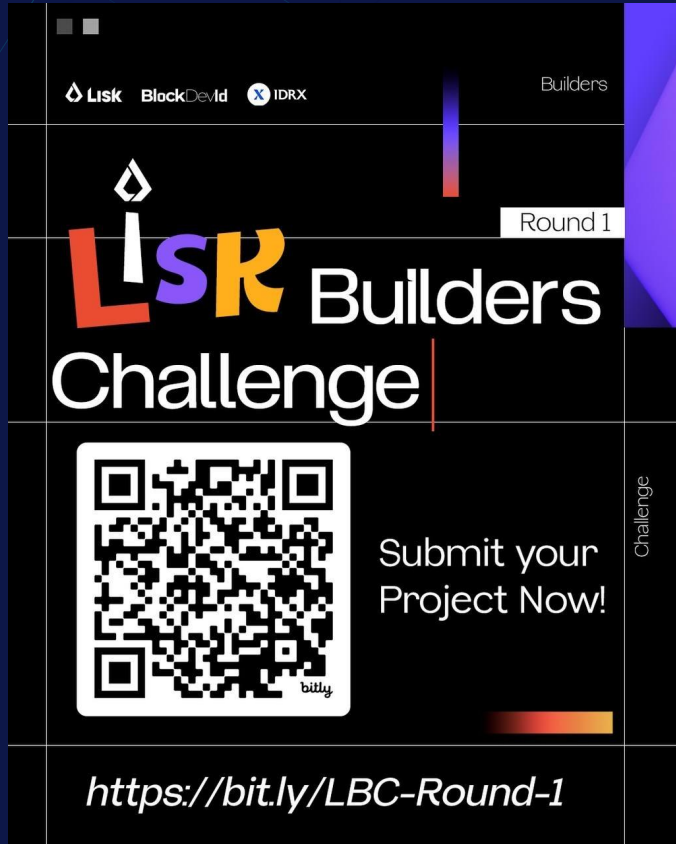
Most Engaging & Creative **3MIO** IDR (187.5 USD/each)

For 2 Winners

Challenge

YESSSS!!





The poster is for the Lisk Builders Challenge Round 1. It features the Lisk logo at the top left, followed by 'BlockDevId' and 'IDRX' logos. The text 'Builders' is in the top right. The main title 'Lisk Builders Challenge' is in the center, with 'Round 1' in a small box to the right. Below the title is a QR code with the text 'Submit your Project Now!' to its right. At the bottom is the URL 'https://bit.ly/LBC-Round-1'. The word 'Challenge' is written vertically on the right side of the poster.

Timeline:

- Hackathon Start: March 8, 2025
- Kickoff AMA: March 12, 2025
- Workshop: March 19 – April 30, 2025
- Submission Deadline: May 17, 2025
- Announcement of Finalists: May 24, 2025
- Finalists Coaching: May 26–30, 2025
- Demo Day (Jakarta – Hybrid): May 31, 2025

Lisk Developer Experience Survey Form



<https://forms.gle/bMj56NQwCci3aSPR8>



B

Pelita Bangsa Academy Resources



Pelita Bangsa Academy Lisk 1000 IGNITE
BlackDevid X IDR X Xellar

* ♦ *

BUILDing with IDRX Stablecoin on Lisk Bootcamp!

* ♦ *

 GRATIS!

LIVE via Zoom

→ Swipe untuk daftar

Pelita Bangsa Academy Resources:

- **LMS (Learning Management System):**
<https://lms.pelitabangsa.co.id/>
(Class Recording, PDF Material & Other Links)
- **Discord Link:**
<https://discord.gg/Jquvw5DqEv>
(Discussion on #lisk channel)
- **Zoom Links on Google Calendar:**
<https://calendar.google.com/calendar/embed?src=0028bfb69c6aa9e82732209167a85aabf5201caae48f1653782c0365b238277%40group.calendar.google.com&ctz=Asia%2FJakarta>



IDRX on Lisk Sepolia Airdrop Form

Fill it out before Thursday, 10 April 2025 | 18:00



<https://forms.gle/t5NSNTdSRWEgsuDMA>

Lisk Sepolia OP Superchain Faucet

Superchain Faucet

Get free testnet tokens for building applications on the Superchain.

Sign in to use the faucet

Anyone can claim 0.1 test ETH every 24 hours, or verify your onchain identity for more tokens. [See details](#)

Sign in

Network

 OP Sepolia	 Base Sepolia
 Unichain Sepolia	 Ink Sepolia
 Minato Sepolia	 Worldchain Sepolia
 Zora Sepolia	 Mode Sepolia
 Lisk Sepolia	 Cyber Sepolia

<https://console.optimism.io/faucet>

Lisk Sepolia PBA Faucet

Lisk Sepolia Testnet Faucet

Request Sepolia ETH for testing on the Lisk Sepolia Testnet

Chain Overview

+ Add to MetaMask

Network Name	Lisk Sepolia
RPC URL	https://rpc.sepolia-api.lisk.com
Chain ID	4202
Currency Symbol	ETH
Block Explorer	https://sepolia-blockscout.lisk.com

Request Testnet ETH

Connected: 0x47331edc7220ad93d62130ce64c10f7166f4c947

Request 0.01 ETH to Connected Wallet

<https://lisk-sepolia-faucet.vercel.app/>

00

About Danantiri

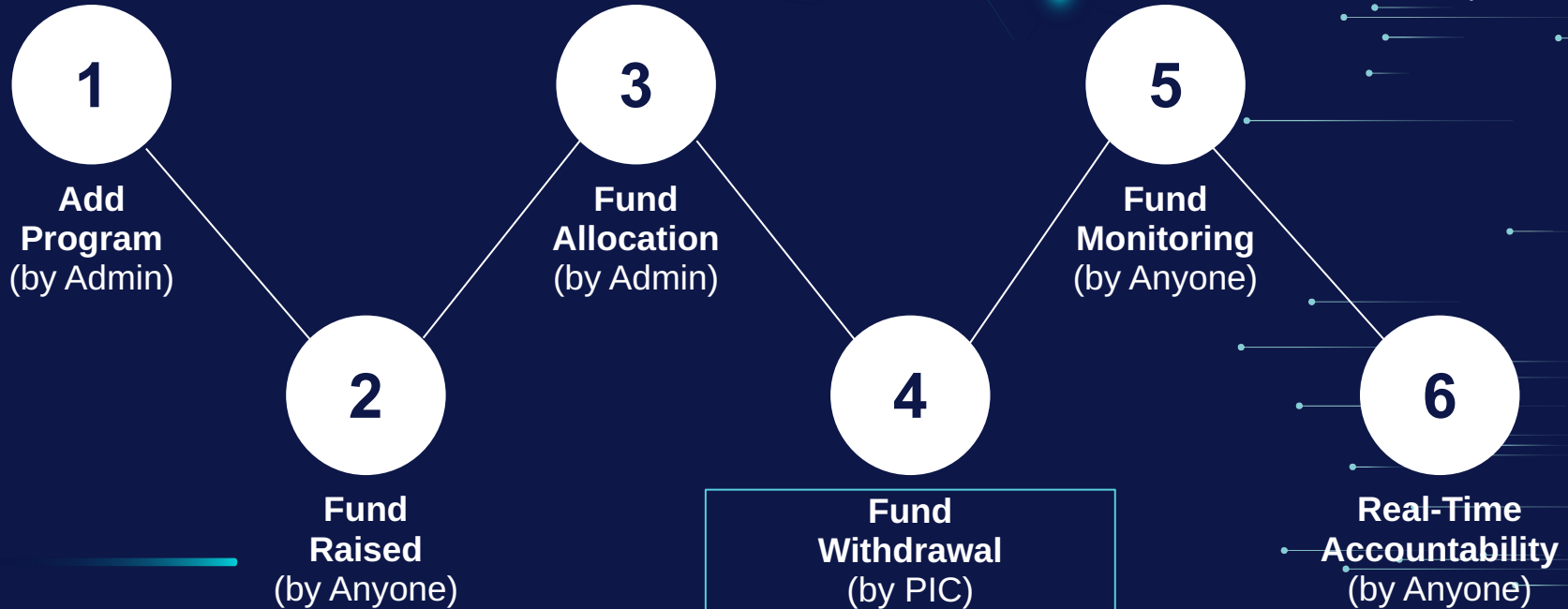
The Problem

- Transparency Issues
- Corruption Risk
- Delayed Reporting

Solution - Danantiri

- Immutable Record
- IDRX stablecoin
- Transparent
- Real-Time Accountability

Business Process



Danantiri | Top Secret



Support Meaningful Projects with Crypto

Your allocated fund will be pooled and distributed to impactful programs.

[Contribute](#)

Managed Fund Amount

12.5 IDR

Programs Funded

4

Featured Programs

MBG

Biar anak sekolah makan enak

0x1RGT7154748V1CE5176479CD0B574AE2448939B5

Testing Program

Testing Program

0x1473289472389578239dfjes

Pembangunan Jalan Tol

Pembangunan Jalan Tol untuk Kontainer

0x3432497DFUOWERN343248098

Pembangunan Sekolah

Pembangunan Sekolah Rakyat

0x3423U43IO24N3B2340934857

Danantiri

Blockchain-Based Government Fund Tracking

Supported Cryptocurrencies

💎 IDRX Lisk Sepolia Testnet

Connect With Us

🗨 Discord PBA

Danantiri | Top Secret

×

Contribute to Danantiri

Enter the amount you want to contribute.

Enter amount in IDR

Approve IDR

Contribute

Danantiri | Top Secret

[Connect Wallet](#)

Admin Panel - Fund Programs Management

Add New Program

[Add Program](#)

Current Programs

MBG

Biar anak sekolah makan enak

0x1RGT7154748V1CE5176479CDG8574AE244B939B5

Testing Program

Testing Program

0x1473289472389678239df/sas

Danantiri | Top Secret

×

MBG

Biar anak sekolah makan enak

PIC Address:
0x1RGT7154748V1CE5176479CDGB574AE244B939B5

Allocate

Remaining Fund:

0 IDR / 25,000,000,000 IDR

Cancel



Pembangunan Sekolah

Pembangunan Sekolah Rakyat

PIC Address: 0x3423U43IO24N3B234G934857

Allocate

Remaining Fund:

100,000,000 IDR / 130,000,000 IDR

Enter amount

Note

Withdraw

Withdraw History

Date	Amount	Note
25-03-2025	30000000	Pembayaran Kontraktor

Danantiri | Top Secret

Pembangunan Sekolah

Pembangunan Sekolah Rakyat

PIC Address: 0x3423U43IO24N3B234G934857

Remaining Fund:

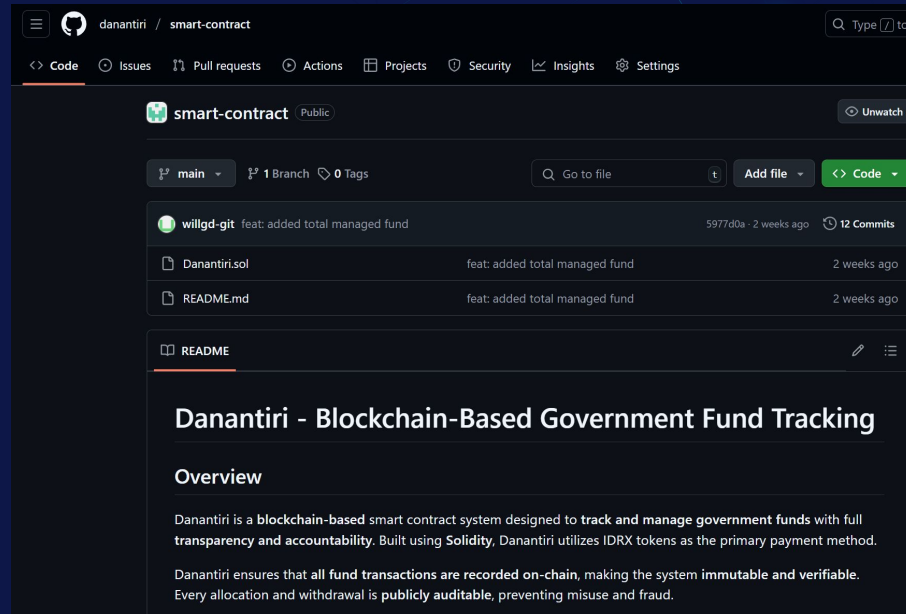


100,000,000 IDR / 130,000,000 IDR

Withdraw History

Date	Amount	Note
30000000	25-03-2025	Pembayaran Kontraktor

GitHub Repo



The screenshot shows the GitHub repository page for 'danantiri / smart-contract'. The repository is public and has 12 commits. The main branch is 'main'. The repository contains files 'Danantiri.sol' and 'README.md', both of which were added 2 weeks ago. The README file is selected, showing the title 'Danantiri - Blockchain-Based Government Fund Tracking' and an 'Overview' section. The overview text states: 'Danantiri is a blockchain-based smart contract system designed to track and manage government funds with full transparency and accountability. Built using Solidity, Danantiri utilizes IDR tokens as the primary payment method. Danantiri ensures that all fund transactions are recorded on-chain, making the system immutable and verifiable. Every allocation and withdrawal is publicly auditable, preventing misuse and fraud.'

<https://github.com/danantiri/smart-contract>

01

Review

What we learned in previous session

1. Smart Contract IDE - Remix
2. Block Explorers - Lisk Blockscout
3. Key Components of Smart Contracts
4. Data Types (Struct, uint, mapping, address, etc)
5. Modifier
6. Event
7. Error Handling
8. Develop Danantiri Smart Contract

Data Types

Signed / Unsigned
Integers (int/uint)

Enums (enum)

Booleans (bool)

Addresses (address)

Bytes (bytes)

Strings (string)

Function Types

	Read	Write	
View			<pre>function getBalance() 317 gas public view returns (uint) { return address(this).balance; }</pre>
Pure			<pre>function addNumbers(uint a, uint b) public pure returns (uint) { return a + b; }</pre>
Non-specified			<pre>function store(uint256 num) public { number = num; }</pre>

Visibilities

Private

Public

Internal

External

02

Token Standards Overview

Cryptocurrency



Cryptos: 2.4M+ Exchanges: 770 Market Cap: \$2.57T ▼ 0.75% 24h Vol: \$87.56B

Implementasi: Stablecoin | Utility | Governance | Security | Equity Tokens

Sumber: coinmarketcap.com

- 1  Bitcoin BTC
- 2  Ethereum ETH
- 3  Tether USDT
- 4  BNB BNB
- 5  Solana SOL

Types of Cryptocurrencies

Coin

- Built into a blockchain - **native to its own blockchain**
- Either **mined** (in PoW) or **earned** (in PoS)



Token (ERC-20)

- Built on top of an existing blockchain - **not native**
- Relatively **easy** to create
- Usually **Distributed** by project owner



NFT (Non Fungible Token) - ERC-721 / ERC-1155



Bored Ape Yacht Club

Market Cap: USD 712.19M



CryptoPunks

Market Cap: USD 2.42B

Sumber: nftvaluations.com

Other Token Standards

ERC-223

<https://ethereum.org/en/developers/docs/standards/tokens/erc-223/>

Soulbound Token

<https://ethereum.org/en/decentralized-identity/#soulbound>

ERC-777

<https://ethereum.org/en/developers/docs/standards/tokens/erc-777/>

ERC-4626

<https://ethereum.org/en/developers/docs/standards/tokens/erc-4626/>

03

What is IDRX?

What is Stablecoin?



A stablecoin is a type of cryptocurrency that is **pegged to a stable asset** (like the US dollar, gold, or a basket of assets) to reduce price volatility

What is IDRX?

IDRX is a blockchain-based digital currency, backed **1:1** by the **Indonesian Rupiah**, designed to offer **stability, transparency, and seamless** interaction with Web3 financial platforms.



Supported Chains



Enables **seamless interaction** with decentralized finance
(DeFi) platforms using IDRX

IDRX on Lisk Sepolia Airdrop Form

Fill it out before Thursday, 10 April 2025 | 18:00



<https://forms.gle/t5NSNTdSRWEgsuDMA>

04

Deep Dive into ERC-20

Analogy

	A	B	C
1	account	balance	
2	0xJason	10,000 IDRX	
3	0xYevon	150,000 IDRX	
4	0xPeter	20,000 IDRX	
5	0xWilliam	1,000,000 IDRX	
6	0xHeinrich	1,500,000 IDRX	
7	0xDeo	1,200,000 IDRX	
8			

< >

IDRX (ERC-20)

Other Smart Contracts

ERC-20

Methods

```
1 function name() public view returns (string)
2 function symbol() public view returns (string)
3 function decimals() public view returns (uint8)
4 function totalSupply() public view returns (uint256)
5 function balanceOf(address _owner) public view returns (uint256 balance)
6 function transfer(address _to, uint256 _value) public returns (bool success)
7 function transferFrom(address _from, address _to, uint256 _value) public returns
8 function approve(address _spender, uint256 _value) public returns (bool success)
9 function allowance(address _owner, address _spender) public view returns (uint256)
```

[Show less](#)[Copy](#)

Events

```
1 event Transfer(address indexed _from, address indexed _to, uint256 _value)
2 event Approval(address indexed _owner, address indexed _spender, uint256 _value)
```

[Copy](#)

OpenZeppelin Contracts Wizard


Forum Docs

Solidity
Cairo
Stellar
Stylus

ERC20
ERC721
ERC1155
Stablecoin*
Real-World Asset*
Governor
Custom

SETTINGS

Name: IDRX
Symbol: IDRX

Premit: 1000000

FEATURES

☒ Mintable
☐ Burnable
☐ Pausable
☐ Callback
☒ Permit
☐ Flash Minting

```

// SPDX-License-Identifier: MIT
// Compatible with OpenZeppelin Contracts ^5.0.0
pragma solidity ^0.8.22;

import {ERC20} from "@openzeppelin/contracts/token/ERC20/ERC20.sol";
import {ERC20Permit} from "@openzeppelin/contracts/token/ERC20/extensions/ERC20Permit.sol";
import {Ownable} from "@openzeppelin/contracts/access/Ownable.sol";

contract IDRX is ERC20, Ownable, ERC20Permit {
    constructor(address recipient, address initialOwner)
        ERC20("IDRX", "IDRX")
        Ownable(initialOwner)
        ERC20Permit("IDRX")
    {
        _mint(recipient, 1000000 * 10 ** decimals());
    }

    function mint(address to, uint256 amount) public onlyOwner {
        _mint(to, amount);
    }
}

```

Deploy

Deployed Contracts 1

IDRX AT 0XAE0...96B8B (MEMC)

Balance: 0 ETH

approve	address spender, uint256 amount
mint	address to, uint256 amount
permit	address owner, address spender, uint256 amount, bool allow
renounceOwn...	
transfer	address to, uint256 value
transferFrom	address from, address to, uint256 value
transferOwne...	address newOwner
allowance	address owner, address spender
balanceOf	address account
decimals	

<https://wizard.openzeppelin.com/>

05

Interact with IDRX Stablecoin to the Smart Contract

IDRX Implementation Adding Interface

IERC20: Widely-used standard contract for creating digital tokens (like IDRX, USDC, etc) on blockchain platforms such as Ethereum or Lisk

Interface:

- Defines the functions your contract can call.
- Ensures communication is correct and secure.

```
interface IERC20 {  
  
    function transferFrom(address sender, address recipient, uint256  
amount) external returns (bool);  
  
    function transfer(address recipient, uint256 amount) external  
returns (bool);  
  
    function balanceOf(address account) external view returns  
(uint256);  
  
}
```


IDRX Implementation Adding State + Constructor

Ensure ERC20 interface is initiated
and saved on the blockchain.

* Need to change the constructor

```
IERC20 public idrxToken;  
  
constructor(address _tokenAddress) {  
    require(_tokenAddress != address(0), "Invalid token address");  
    owner = msg.sender;  
    idrxToken = IERC20(_tokenAddress);  
}
```

IDRX Implementation Adding Function Send Fund

Send Fund:

Allows end-user to send IDRX to this smart contract.

```
function sendFund(uint256 amount) external {  
    ...  
}
```

IDRX Implementation Adding Function Allocate Fund

Allocate Fund:

Allows admin to allocate collected fund from the end-user to the registered program.

```
function allocateFund(uint256 _programId) external onlyAdmin {  
    ...  
}
```

IDRX Implementation Adding Function Withdraw Fund

Withdraw Fund:

Allows **PIC** to withdraw allocated fund by the admin. This function also record how and how much funds are used.

```
function withdrawFund(uint256 _programId, string calldata _history,  
uint256 _amount) external onlyPIC(_programId) {  
    ...  
}
```

IDRX Implementation

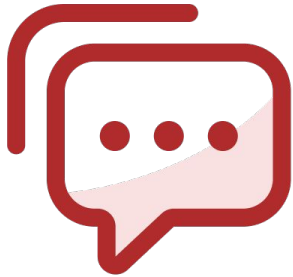
Adding Function Get Program History

Get Program History:

Retrieve all history based on which
parameter program ID sent.

```
function getProgramHistory(uint256 _programId) external view returns  
(History[] memory) {  
    ...  
}
```

Q & A



Audience Q&A

① Click **Present with Slido** or install our
while presenting.

to show live Q&A

Wrap Up

Share 3 things you learn today

On the Zoom Chat

Foto Bersama

**Pelita Bangsa
Academy**

**Mall of Indonesia, Lantai 2, Unit 2F-A7B,
Jalan Boulevard Barat Raya No. 1,
Kelapa Gading Jakarta Utara 14240.**



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